

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Poojasree B (192411091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Poojasree B

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ARTIFICIAL INTELLIGENCE

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CSAIT10

A cleaning robot is placed in an unknown building with no map available. It can only sense its immediate surroundings. Explain how an Online Search Agent would explore this environment, and discuss the challenges of searching with partial/no prior information compared to offline search.

A cleaning robot placed in an unknown building has no prior map. It can only sense its immediate surroundings using sensors such as camera, ultrasonic sensors, or obstacle detectors. Therefore, it must explore the environment while simultaneously building knowledge about it.

Working of the Online Search Agent

1. Initial state

The robot starts at an unknown location with no map.

2. Perceive the Environment

It senses nearby obstacles, rooms, corridors, walls, and unvisited areas.

3. Choose an Action:

Based on its current knowledge, it selects a direction to move, such as:

- * Move forward
- * Turn left or right
- * Enter a new room
- * Return to a previously visited location

4. Update Knowledge:

After taking an action, the robot observes the results and updates its internal memory (map).

5. Explore New Areas:

It gives preference to unexplored locations until the building is completely explored and cleaned.

For example, if the robot reaches a junction it may choose one corridor, remember the junction, and return to explore the other corridor.

⇒ So it must explore the environment while simultaneously building knowledge about it.

Challenges

Online search	Offline search
* The environment is initially unknown * The agent must explore while searching * Actions may have unexpected results * It may need to backtrack. * Search can be slower. * It must handle incomplete info.	* The complete environment is known. * The complete path can be planned beforehand. * The result of actions is generally known. * The optimal path can be calculated directly * Search is usually more efficient. * It has complete prior info.

Conclusion:

An online search agent is suitable for the cleaning robot because it can make decision based on current perceptions and gradually learn the structure of the unknown building.

2. Smart Thermostat Agents

A. Simple Reflex Agent

A Simple Reflex Agent makes decisions only using the current temperature reading and the target temperature

Rules

- * If current temperature $<$ target temperature $\rightarrow$ Turn ON heater
- * If current temperature $>$ target temperature $\rightarrow$ Turn ON cooler
- * If current temperature $=$ target temperature $\rightarrow$ Turn OFF heater and cooler

Limitation

The simple reflex agent only considers the current temperature. It does not remember:

- * Previous temperature
- * Whether the heater was recently turned on
- * The rate at which the room temperature is changing.

* Heat loss from the room.

* Whether doors or windows are open

B Model-Based Reflex Agent

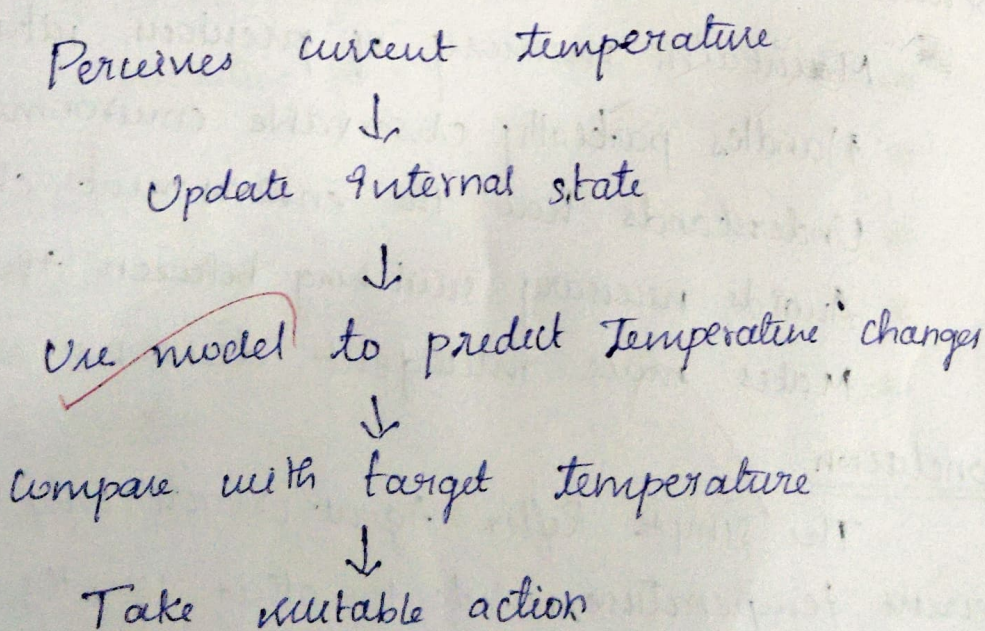
A model-Based Reflex Agent maintains an internal state of the environment.

Internal state to maintain

The thermostat should maintain:

1. Current room temperature
2. Previous temperature
3. Target temperature
4. Temperature trend
5. Current heating/cooling status
6. Estimated room conditions

Working



Example:

Suppose:

- * Target temperature = 24°C
- * Current temperature = 23°C
- * Temperature is rapidly increasing

A simple reflex agent immediately turns ON the heater.

However, a model-based agent understands that the temperature is already increasing and may reach 24°C soon. Therefore, it may keep the heater OFF.

Why Model-Based Agent Performs Better

The Model-Based Reflex Agent performs better because it:

- * Maintains memory of previous situations
- * Handles partially observable environments
- * Understands how the environment changes
- * Avoids unnecessary switching between them
- * Makes more intelligent decisions

Conclusion

The Simple Reflex Agent reacts only to the current temperature, while the other uses the current perception.

Self-Driving Car on City Roads

A. Percepts

A self-driving car receives information from its sensors.

Main percepts

- * Other vehicles
- * Pedestrians
- * Traffic lights
- * Traffic signs
- * Road lanes and markings
- * Obstacles
- * Road conditions
- * GPS location
- * Speed of nearby vehicles
- * Distance from other objects
- Weather and visibility conditions

Sensor used

- * Cameras
- * LIDAR
- * Radar
- * Ultrasonic sensors
- * GPS
- * IMU sensors

B. Possible Actions

The car can perform the following actions:

- * Accelerate
- * Brake
- * Stop
- * Turn left
- * Turn right
- * Change lanes
- * Maintain speed
- * Overtake
- * Follow another vehicle
- * Park
- * Take an alternate route

C. Characterization of the Environment

1. Partially Observable

The environment is partially observable.
The car cannot observe everything at all times because:

- * Other vehicles may block its view
- * Objects may be hidden behind buildings
- * Sensors have limitations.

2. Stochastic

The environment is stochastic. The result of an action is not always predictable.

For example:

- * A pedestrian may suddenly cross the road
- * Another car may suddenly change lanes
- * Traffic conditions may change
- * A vehicle may unexpectedly stop

Therefore, the car must handle uncertainty.

3. Dynamic

The environment is dynamic. The environment changes continuously even while the car is deciding what to do.

For example:

- * Vehicles move
- * Pedestrians walk
- * Traffic light change
- * Traffic conditions change

4. Continuous

The environment is mainly continuous.

Examples:

- * Speed can have many values
- * Steering angle can vary continuously
- * Position changes continuously
- * Time is continuous.

Although some events such as traffic-light states are discrete, the overall environment is considered continuous.

D. Most Suitable Agent Architecture

The most suitable architecture is a Utility-Based Agent with Model-Based and Goal-Based capabilities.

Why?

A self driving car must:

- * Maintain an internal model of the environment.

- * Achieve goals such as reaching the destination

- * Choose the safest and most efficient action

- * Balance multiple factors.

Utility Factors:

The car may consider:

- * Safety
- * Travel time
- * Fuel or energy consumption
- * Passenger comfort
- * Traffic conditions
- * legal traffic rules

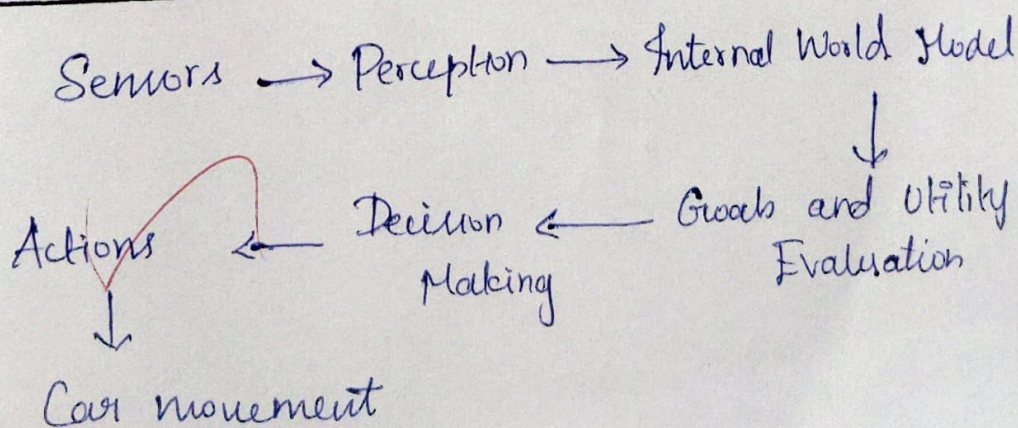
For example, between two routes:

* Route A is shorter but highly congested.

* Route B is slightly longer but safer and less congested.

A utility-based agent can evaluate both routes and choose the one with the highest overall utility.

Architecture



Conclusion:

A Utility-Based Agent is the most suitable architecture for a self-driving car because the car must operate in a partially observable, stochastic, dynamic, and continuous environment.

It can use its internal model to understand the environment, achieve goals, and select the safest and most beneficial action among several alternatives.